Stakeholder Engagement Plan: Optimising RAG Document Retrieval for Agronomic Advice

1. Stakeholder Groups

Group 1 - Smallholder farmers (end users). These are the people who will ask questions of the system and act on what it surfaces. They are central because the entire project exists to serve their information needs at the moment farming decisions are made. What matters most to them: getting accurate, trustworthy guidance quickly; being understood even when they ask short, informal, or local-language questions; and advice that fits their actual crops and conditions rather than only major staples.

Group 2 - Agricultural extension officers and agronomists (domain experts / intermediaries). These are the trained advisers who support farmers and hold the expertise to judge whether a retrieved document truly answers a question. They are critical both as intermediaries who will use the system in the field and as the source of reliable relevance judgements. What matters most to them: correctness and safety of advice, keeping their professional expertise central rather than displaced, and being able to trace and verify what the system surfaces.

2. Potential Benefits and Risks

Smallholder farmers.

- ✓ *Benefits:* faster access to relevant guidance, better diagnoses, less wasted spending on wrong inputs, greater resilience to pests and climate shocks, and information reaching those with limited access to specialists.
- ✓ *Harms/concerns:* being misled by a plausible-but-wrong document (a hard negative), exclusion if their crop or dialect is under-represented, and over-reliance on the system without recourse to human advice.
- ✓ *Considerations:* social and economic - a wrong recommendation carries direct financial cost; cultural and linguistic - farmers phrase questions in local languages and idioms.

Extension officers and agronomists.

- ✓ *Benefits:* a tool that reduces search time across scattered factsheets, supports consistent advice, and lets them focus expertise where it matters most.
- ✓ *Harms/concerns:* erosion or de-skilling of their professional role, automation bias among farmers, and reputational risk if the system surfaces incorrect guidance under their name.
- ✓ *Considerations:* ethical - accountability for advice; professional - trust that the tool supports rather than replaces judgement.

3. Engagement Method

Smallholder farmers -> Distributed Dialogue. Distributed Dialogue is appropriate because farmers are geographically dispersed across rural areas, often cannot travel to a central venue, and vary widely in crop, language, and literacy. This method runs structured conversations across multiple local sites and channels, meeting farmers where they are rather than requiring them to come to us. It is suitable for this group because it surfaces the real language, phrasing, and question types farmers actually use - exactly the informal and local-language queries our system must handle - and it reaches voices that a single centralised workshop would exclude (minor-crop growers, women farmers, lower-literacy participants). It would be conducted through facilitated sessions at local cooperatives, markets, and via community radio or mobile channels, in local languages, with facilitators collecting the actual questions farmers ask and their reactions to sample retrieved documents. This gives us authentic query data and direct signal on whether surfaced guidance feels relevant and trustworthy to the people who will act on it.

Extension officers and agronomists -> Participatory Design. Participatory Design (co-design workshops) is appropriate because this group holds the domain expertise the system depends on, and involving them directly turns them from subjects into co-creators. It is suitable because agronomists can contribute and review relevance judgements, flag hard negatives that look right but are dangerous, and shape how results are presented so the tool supports rather than displaces their role. It would be conducted through iterative workshops where officers contribute real farmer questions, label and rank candidate documents, review the system's top-5 outputs against their own judgement, and help define what "correct" retrieval looks like across crops. Their sustained involvement builds trust, improves label quality, and keeps professional accountability central.

4. Influence on Project Design

Feedback from these groups would directly reshape the project. Distributed Dialogue with farmers would expand and diversify our dataset with authentic informal and local-language queries, improving coverage of under-represented crops and phrasings, and informing consent practices for any real questions collected. Participatory Design with agronomists would raise labelling quality and sharpen how we identify hard negatives, directly improving evaluation (per-crop, intent-aware relevance). Together they would refine outputs (how results are presented and sourced), guide model design toward genuine semantic intent over keyword overlap, and shape deployment decisions - positioning the system as a traceable decision aid that keeps human expertise in the loop.